AUDIENCE SEGMENTATION & FAN STRATEGY

# Who is ready to discover a new artist?

I analyzed historical Last.fm listening behavior to find audiences a record label could test with different discovery and catalog campaigns.

| 19.1M | 992 | 926 |
|---|---|---|
| **Parsed listening events** | **Source users** | **Eligible listeners** |

## Two listener profiles

| | Focused | Exploratory Power |
|---|---|---|
| Listeners | **454 (49%)** | **472 (51%)** |
| Plays per observation day | **9.67** | **42.19** |
| Artist breadth per observation day | **0.62** | **2.49** |
| Share of plays for top artist | **8.67%** | **5.92%** |
| Active-day rate | **28.22%** | **71.36%** |

**WHAT CHANGED WITH ENGAGEMENT**

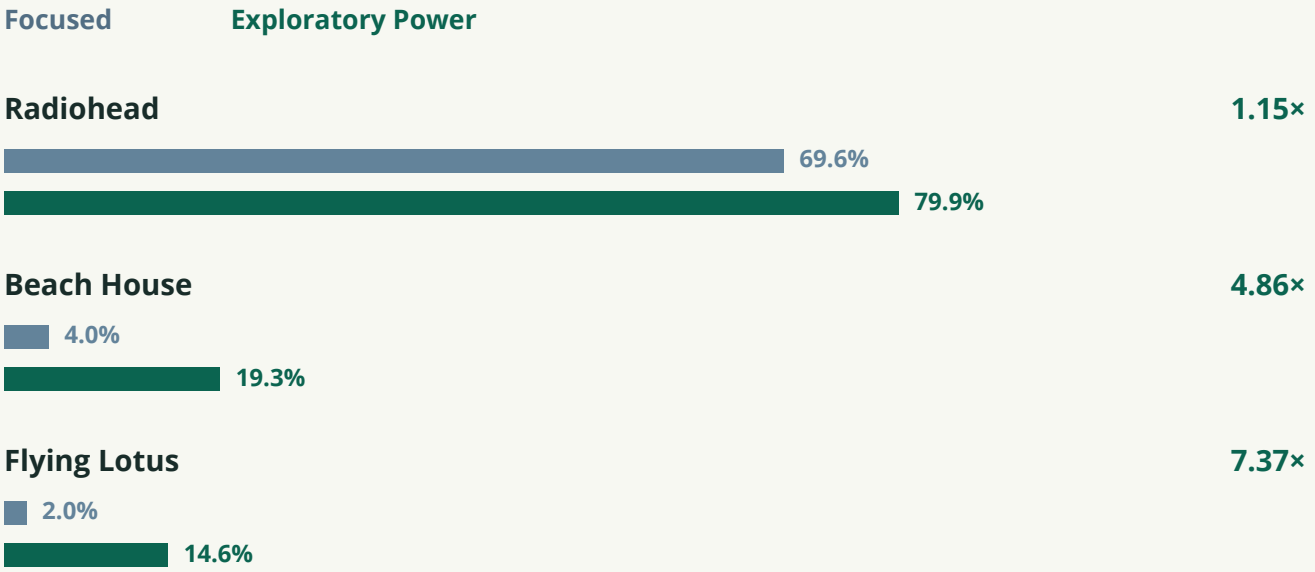
Exploratory Power Listeners played music about 4.4 times as often and showed about four times the artist breadth. Their top artist accounted for a smaller share of listening.

ARTIST REACH & AFFINITY

# Who reaches which fans?

Radiohead reached 693 listeners across the cohort and over-indexed at 1.15×. Beach House reached fewer listeners overall, with nearly five times the penetration in the Exploratory Power segment.

## Listener penetration by segment




Penetration is the share of a segment that played an artist at least once. Over-index divides Exploratory Power penetration by Focused penetration. The discovery shortlist requires at least 20 listeners overall, five in each segment, and 10% Exploratory Power penetration.

**CAMPAIGN TESTS**

For Exploratory Power Listeners, use an adjacent artist to test discovery and measure first-time plays. For Focused Listeners, lead with a familiar artist and measure repeat catalog plays. Compare both treatments with holdout groups.

METHODS & VALIDATION

# How I tested the segments

## Data and eligibility

The historical Last.fm 1K cohort spans roughly 2005–2009. I required at least 500 plays and normalized time-dependent features by each listener's inclusive UTC calendar-date observation window. Artist breadth is distinct artist names across the window divided by observation days.

## Model selection

I log-transformed listening intensity, artist breadth and top-artist concentration, then standardized them alongside active-day rate. I compared K-Means solutions from two to eight clusters. The two-cluster model scored highest on silhouette at 0.3323.

## Threshold sensitivity

Increasing the minimum to 1,000 plays preserved segment assignments for 870 of the 890 overlapping listeners (97.75%). Different random starts also reproduced the same 454 / 472 split and silhouette score in the original analysis.

## What the results can support

The historical cohort and modest cluster separation make these profiles directional. Artist names can be inconsistent, and the data cannot establish why listeners behaved as they did. Current first-party data and controlled campaigns would be needed to validate the recommendations.

DATA SOURCE

Last.fm 1K User Dataset · Òscar Celma